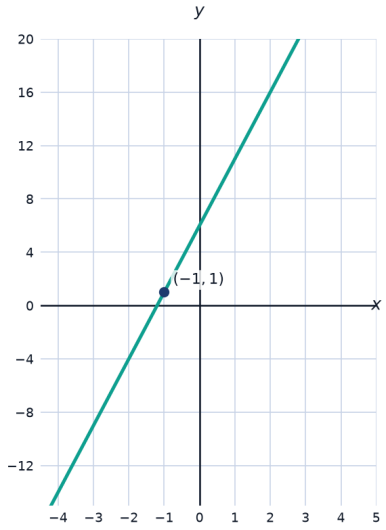
# Exponential & Logarithmic Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

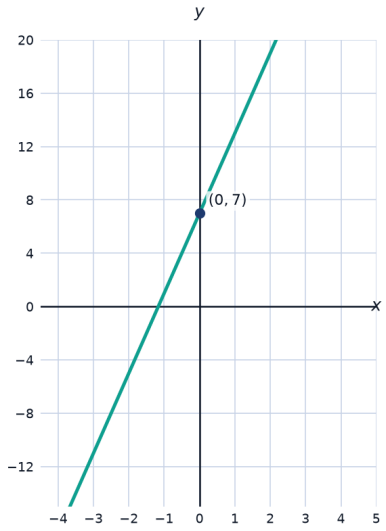
**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(5)$.




**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(2)$.



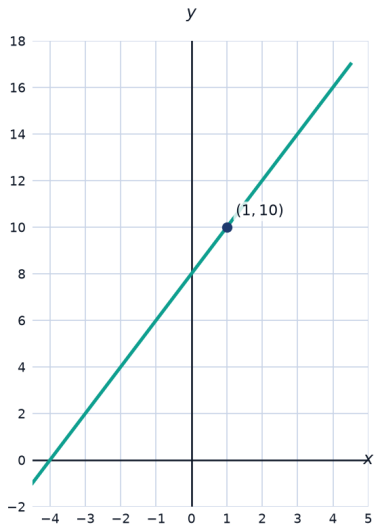

**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for solving equations with rational expressions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

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**4** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(2)$.

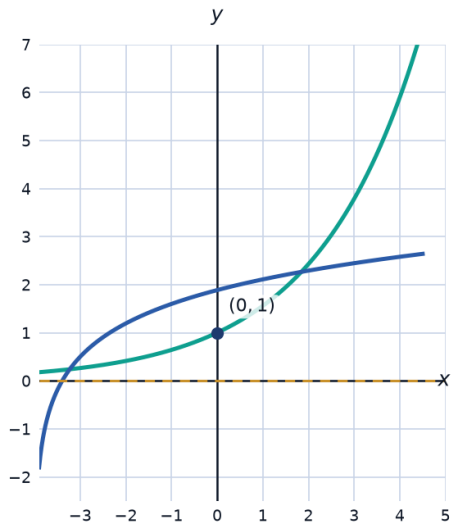



**5** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for graphical and analytical solutions to rational equations consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

**6** [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the sum of all real solutions to the equation $3^x + 3^{3-x} - 10 = |3^{x-1} - 3|$ ? Show your complete algebraic work.




**7** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · AdvancedExperimental laboratory data pairs $(x, y)$ are transformed by computing $Y = \ln(y)$. A linear regression performed on $(x, Y)$ yields the best-fit line $Y = -0.42x + 2.85$. Write the corresponding non-linear model for $y$ as an exponential function of $x$ in the form $y = ab^x$, giving $a$ and $b$ to three decimal places. Then predict the value of $y$ when $x = 5$.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

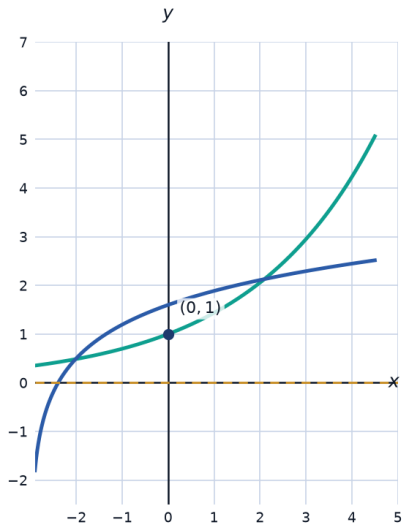
Level 1 · FoundationsA sample of a radioactive isotope has an initial mass of 250 mg and decays at a continuous rate of 4.8% per day according to $M(t) = 250e^{-0.048t}$, where $t$ is measured in days. Determine the exact daily multiplicative decay factor $b$ in the form $M(t) = 250b^t$, and calculate the daily percentage loss rounded to two decimal places.

**9** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A culture of bacteria doubles in population every 3 hours. If the initial count is 250 bacteria, write an exponential model $B(t)$ for the population after $t$ hours, and find the population after 12 hours.

**10** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Unit 2 · Exponential and Logarithmic Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.




**11** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

**12** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A radioactive substance follows the decay model $M(t) = M_0e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant $k$ and the substance's half-life in days to the nearest tenth.

**13** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

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**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · MediumFind all real solutions to the trigonometric equation $2\cos^2\theta + 3\sin\theta - 3 = 0$ on the interval $[0, 2\pi)$.

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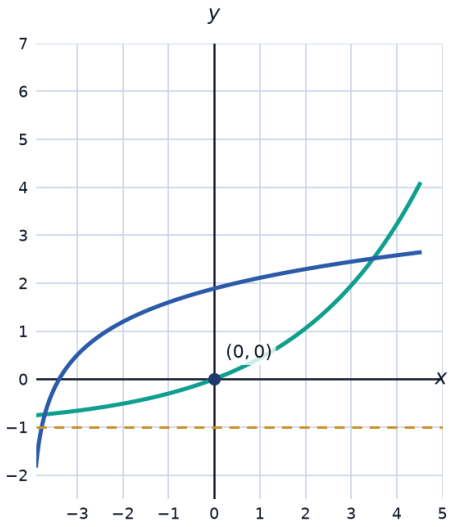
## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions & Their Graphs

Level 1 · FoundationsFind the exact value of $\tan\left(\frac{11\pi}{6}\right)$ and $\sec\left(\frac{11\pi}{6}\right)$ using reference angles and unit circle coordinates.

**16** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Matrices & Systems

Level 3 · AdvancedFind all values of $\theta$ in $[0, 2\pi)$ for which the rose curve $r = 4\sin(3\theta)$ passes through the pole (origin), and find the maximum distance of any point on the curve from the origin.




**17** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Probability & Statistics

Let $f(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$. Identify the locations of all vertical asymptotes and removable discontinuities (holes) of $f(x)$.

**18** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

A bivariate dataset $(x, y)$ is transformed into a semi-log plot $(x, \ln y)$, yielding the linear regression model $\ln(\hat{y}) = 1.5 + 0.4x$. Express $\hat{y}$ as an exponential function of $x$ in the form $\hat{y} = a \cdot b^x$, and find the predicted percentage growth rate per unit increase in $x$.

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**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Polynomial Functions

Solve the logarithmic equation $\log_2(x+3) + \log_2(x-3) = 4$ for all real solutions $x$. State any extraneous roots.

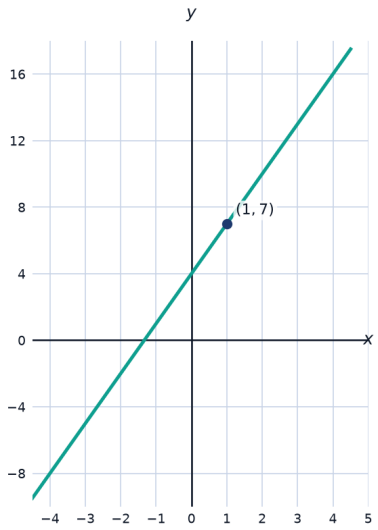
**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Rational Expressions

A projectile is launched from ground level with height modelled by $h(t) = -16t^2 + 64t + 80$ feet after $t$ seconds. Find (a) the maximum height reached, and (b) the time when it hits the ground.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(2)$.




**a.** 10

**b.** 13

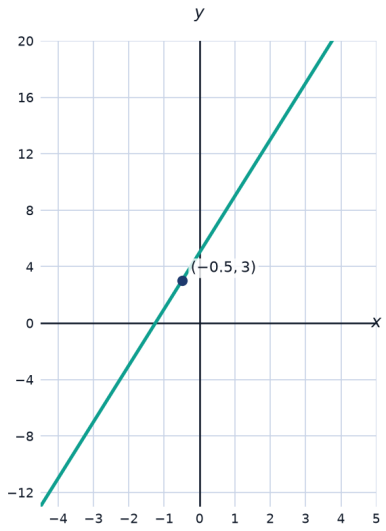
**c.** 6

**d.** 14

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**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in adding and subtracting rational expressions, evaluate $f(2)$.




**a.** 13

**b.** 17

**c.** 8

**d.** 17

**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for simplifying rational expressions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

**a.** 14.5

**b.** 15.5

**c.** 16.5

**d.** 18.5

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for adding and subtracting rational expressions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** 15.5

**b.** 16.5

**c.** 17.5

**d.** 19.5

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**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for multiplying and dividing rational expressions consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

**a.** 16.5

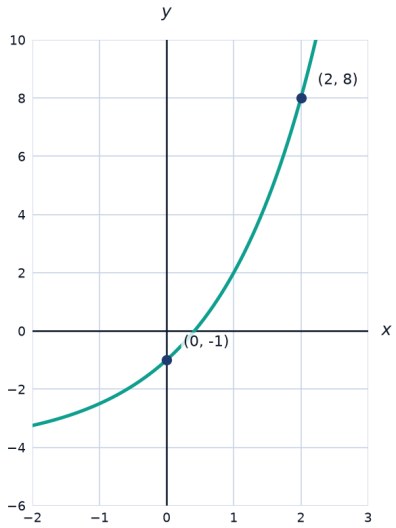
**b.** 17.5

**c.** 18.5

**d.** 20.5

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the $xy$-plane, where $a$, $b$, and $c$ are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?




**a.** $-\frac{5}{2}$

**b.** $-\frac{7}{2}$

**c.** $-\frac{13}{4}$

**d.** $\frac{3}{2}$

**27** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The function $f$ is defined by $f(x) = a \cdot b^x + c$, where $a$, $b$, and $c$ are constants with $a > 0$, $b > 0$, and $beq1$. In the $xy$-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

**a.** 20

**b.** 24

**c.** 32

**d.** 44

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**28** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of $a$.

**a.** 2

**b.** 3

**c.** 4

**d.** 6

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the value of $y$ when $x = 3$ for the function $f(x) = 2^{x+1}$?

**a.** 4

**b.** 8

**c.** 6

**d.** 16

**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the half-life formula for exponential decay?

**a.** $A(t) = A_0(2)^{t/h}$

**b.** $A(t) = A_0 e^{-t}$

**c.** $A(t) = A_0/2^t$

**d.** $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve: $e^{2x} = 5$. Express in terms of $\ln$.

**a.** $x = \ln k**2 + 1 - k$

**b.** $x = \ln \frac{k**2+1}{k}$

**c.** $x = \frac{\ln k**2+1}{k}$

**d.** $x = \frac{k}{\ln k**2+1}$

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**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

**a.** $A = Pe^{rt}$

**b.** $A = P/e^{rt}$

**c.** $A = P(1 + r)^t$

**d.** $A = Prt$

**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The equation $\log_a x = b$ and $\log_b x = a$ simultaneously implies:

**a.** No simultaneous solution

**b.** $a = b$

**c.** $x = 0$

**d.** $x = a^b = b^a$

**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve: $5^x = 12$. Give exact answer.

**a.** $x = 12/5$

**b.** $x = \frac{\ln 12}{\ln 5}$

**c.** $x = \ln 12 - \ln 5$

**d.** $x = \log_{12} 5$

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Write using natural log: $\log_b x =$

**a.** $\frac{\ln b}{\ln x}$

**b.** $\ln x - \ln b$

**c.** $\ln x \cdot \ln b$

**d.** $\frac{\ln x}{\ln b}$

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The value of $e$ to 4 decimal places is:

**a.** 2.1787

**b.** 3.1416

**c.** 2.6180

**d.** 2.7183

**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which is the inverse of $f(x) = \log_2 x$?

**a.** $g(x) = \log_2 x$

**b.** $g(x) = 2x$

**c.** $g(x) = x^2$

**d.** $g(x) = 2^x$

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Convert to exponential form: $\log_5 25 = 2$.

**a.** $5^{25} = 2$

**b.** $2^5 = 25$

**c.** $5^2 = 25$

**d.** $25^2 = 5$

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Polynomial Functions

Which is the y-intercept of $f(x) = 5 \cdot 3^x$?

**a.** $(1, 15)$

**b.** $(0, 15)$

**c.** $(0, 5)$

**d.** $(0, 3)$

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Rational Expressions

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find $A/A_0$ at $t = 10$.

**a.** 8

**b.** 2

**c.** 4

**d.** 10

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Which of these is equivalent to $e^{2\ln 3}$?

**a.** $2e^3$

**b.** 9

**c.** 6

**d.** $\ln 9$

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Sequences & Series

Find the derivative of $f(x) = x^x$.

**a.** $x^x \ln x$

**b.** $x^{x+1}$

**c.** $x \cdot x^{x-1}$

**d.** $x^x(1 + \ln x)$

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**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Conic Sections

Find the number of solutions to $e^x = x + 2$.

**a.** 1

**b.** 2

**c.** Infinitely many

**d.** 0

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Trigonometric Functions

Solve: $3^x = 2^{x+1}$. Use logarithms.

**a.** $x = \frac{\ln 2}{\ln 3}$

**b.** $x = 1$

**c.** $x = \frac{\ln 3}{\ln 2}$

**d.** $x = \frac{\ln 2}{\ln 3 - \ln 2}$

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Functions & Their Graphs

Show that $2^{10} = 1024 \approx 10^3$. Use this to approximate $\log_{10} 2$.

**a.** $\log_{10} 2 \approx 3/10 = 0.3$

**b.** $\log_{10} 2 = 1$

**c.** $\log_{10} 2 = 1/2$

**d.** $\log_{10} 2 = 0.5$

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